Sequences & Series

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Arithmetic sequences**

For the arithmetic sequence with $a_1 = 9$ and common difference $d = 6$, find a_{10}.

$a_{10} = a_1 + (10 - 1)d = 9 + 9(6) = 63.$

2 **[Foundational]** • **Target: Arithmetic sequences**

For the arithmetic sequence with $a_1 = 15$ and $d = -4$, find a_{11}.

$a_{11} = 15 + 10(-4) = 15 - 40 = -25.$

3 **[Foundational]** • **Target: Arithmetic series**

Find S_{10} for the arithmetic sequence with $a_1 = 9$, $d = 6$ (using $a_{10} = 63$ from above).

$S_{10} = \frac{n(a_1 + a_n)}{2} = \frac{10(9 + 63)}{2} = 5(72) = 360.$

4 **[Foundational]** • **Target: Geometric sequences**

For the geometric sequence with $a_1 = 6$ and common ratio $r = 2$, find a_8.

$a_8 = a_1 r^7 = 6(2^7) = 6(128) = 768.$

5 **[Foundational]** • **Target: Geometric sequences**

A geometric sequence has $a_1 = 2$ and common ratio $r = 5$. Find the smallest n for which $a_n = 1250$.

$a_n = 2 \cdot 5^{n-1}$. Setting $2 \cdot 5^{n-1} = 1250$ gives $5^{n-1} = 625 = 5^4$, so $n - 1 = 4$ and $n = 5$.

6 **[Foundational]** • **Target: Geometric series (finite)**

Find S_8 for the geometric sequence with $a_1 = 6$, $r = 2$ (using $a_8 = 768$ from above).

$S_8 = \frac{a_1(r^8 - 1)}{r - 1} = \frac{6(256 - 1)}{1} = 6(255) = 1530.$

7 **[Foundational]** • **Target: Infinite geometric series**

Find the sum of the infinite geometric series $36 + 12 + 4 + \cdots$.

$a = 36$, $r = \frac{1}{3}$, and $|r| < 1$, so the sum is $\frac{36}{1 - \frac{1}{3}} = \frac{36}{\frac{2}{3}} = 54.$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Infinite geometric series

Find the sum of the infinite geometric series $12 - 4 + \frac{4}{3} - \cdots$.

$a = 12$, $r = -\frac{1}{3}$, and $|r| < 1$, so the sum is $\frac{12}{1-\left(-\frac{1}{3}\right)} = \frac{12}{\frac{4}{3}} = 9$.

9 [Intermediate] • Target: Mathematical Induction

A geometric sequence has its first term a_1 and a common ratio r. The sum of the first three terms is 26, and the sum of the first six terms is 728. If $a_1 = 2$, find the value of r.

Let the first term of the geometric sequence be a_1 and the common ratio be r. The sum of the first n terms of a geometric sequence is given by $S_n = a_1\frac{1-r^n}{1-r}$ (for $req1$). We are given that $a_1 = 2$. The sum of the first three terms is $S_3 = 26$. So, $2\frac{1-r^3}{1-r} = 26$. This simplifies to $\frac{1-r^3}{1-r} = 13$. Using the difference of cubes formula, $1 - r^3 = (1 - r)(1 + r + r^2)$. Thus, $1 + r + r^2 = 13$. Rearranging this, we get a quadratic equation: $r^2 + r - 12 = 0$. Factoring the quadratic, we have $(r + 4)(r - 3) = 0$. This gives possible values for r as $r = 3$ or $r = -4$. Now consider the sum of the first six terms, $S_6 = 728$. So, $2\frac{1-r^6}{1-r} = 728$. This simplifies to $\frac{1-r^6}{1-r} = 364$. We can rewrite $1 - r^6$ as $(1 - r^3)(1 + r^3)$. So, $\frac{(1-r^3)(1+r^3)}{1-r} = 364$. We know that $\frac{1-r^3}{1-r} = 13$. Substituting this into the equation, we get $13(1 + r^3) = 364$. Dividing by 13, we have $1 + r^3 = \frac{364}{13} = 28$. So, $r^3 = 27$. This implies $r = 3$. Both conditions are satisfied when $r = 3$. If we had chosen $r = -4$ from the first condition, $S_6 = 2\frac{1-(-4)^6}{1-(-4)} = 2\frac{1-4096}{5} = 2\frac{-4095}{5} = 2(-819) = -1638$, which is not 728. Therefore, the common ratio is 3.

10 [Intermediate] • Target: Arithmetic sequences

A sequence is defined by the recurrence relation $a_{n+1} = 2a_n - 3$ for $n \geq 1$, with $a_1 = 5$. Consider a second sequence, b_n, where $b_n = a_n - k$ for some constant k. If the sequence b_n is a geometric sequence, find the value of k and the common ratio of the sequence b_n.

The first sequence is defined by $a_{n+1} = 2a_n - 3$, with $a_1 = 5$. We are given a second sequence $b_n = a_n - k$. For b_n to be a geometric sequence, the ratio $\frac{b_{n+1}}{b_n}$ must be constant. Substituting $a_n = b_n + k$ into the recurrence relation for a_n: $b_{n+1} + k = 2(b_n + k) - 3$ $b_{n+1} + k = 2b_n + 2k - 3$ $b_{n+1} = 2b_n + k - 3$ For b_n to be a geometric sequence with common ratio r, we must have $b_{n+1} = rb_n$. Comparing this with the derived equation, we see that r must be 2, and the constant term $k - 3$ must be zero. So, $k - 3 = 0$, which implies $k = 3$. The common ratio r is 2. Let's verify this. If $k = 3$, then $b_n = a_n - 3$. $a_1 = 5$, so $b_1 = 5 - 3 = 2$. $a_2 = 2a_1 - 3 = 2(5) - 3 = 10 - 3 = 7$. So $b_2 = 7 - 3 = 4$. $a_3 = 2a_2 - 3 = 2(7) - 3 = 14 - 3 = 11$. So $b_3 = 11 - 3 = 8$. The sequence b_n starts with 2, 4, 8, ... This is a geometric sequence with first term 2 and common ratio $\frac{4}{2} = 2$ or $\frac{8}{4} = 2$. Thus, $k = 3$ and the common ratio $r = 2$.

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11 **[Intermediate]** • **Target: Arithmetic series**

Consider the function $f(x) = x^3 - 6x^2 + 9x$. Find the area of the region bounded by the graph of $f(x)$ and the x-axis.

First, find the x-intercepts of the function by setting $f(x) = 0$: $x^3 - 6x^2 + 9x = 0$ $x(x^2 - 6x + 9) = 0$ $x(x-3)^2 = 0$ The x-intercepts are $x = 0$ and $x = 3$ (a repeated root, indicating the graph touches the x-axis at $x = 3$ without crossing). To find the area, we need to integrate the function between these intercepts. We need to determine if the function is above or below the x-axis in the interval $(0, 3)$. Let's test a point, say $x = 1$: $f(1) = 1^3 - 6(1)^2 + 9(1) = 1 - 6 + 9 = 4$. Since $f(1) > 0$, the function is above the x-axis in the interval $(0, 3)$. The area is given by the definite integral: Area = $\int_0^3 (x^3 - 6x^2 + 9x)dx$ Now, evaluate the integral: $\int (x^3 - 6x^2 + 9x)dx = \frac{x^4}{4} - 6\frac{x^3}{3} + 9\frac{x^2}{2} = \frac{x^4}{4} - 2x^3 + \frac{9}{2}x^2$ Evaluate the definite integral from 0 to 3: $\left[\frac{x^4}{4} - 2x^3 + \frac{9}{2}x^2\right]_0^3 = \left(\frac{3^4}{4} - 2(3)^3 + \frac{9}{2}(3)^2\right) - \left(\frac{0^4}{4} - 2(0)^3 + \frac{9}{2}(0)^2\right)$ $= \frac{81}{4} - 2(27) + \frac{9}{2}(9) - 0 = \frac{81}{4} - 54 + \frac{81}{2}$ To combine these, find a common denominator, which is 4: $= \frac{81}{4} - \frac{54 \times 4}{4} + \frac{81 \times 2}{4} = \frac{81}{4} - \frac{216}{4} + \frac{162}{4} = \frac{81 - 216 + 162}{4} = \frac{243 - 216}{4} = \frac{27}{4}$ Thus, the area of the region bounded by the graph of $f(x)$ and the x-axis is $\frac{27}{4}$.

12 **[Intermediate]** • **Target: Geometric sequences**

The graph of the function $f(x) = x^2 - 6x + k$ is shown below. The graph has a vertex at $(3, -5)$. Determine the value of the parameter k.

The vertex of a parabola in the form $f(x) = ax^2 + bx + c$ occurs at $x = -b/(2a)$. For $f(x) = x^2 - 6x + k$, we have $a = 1$ and $b = -6$. The x-coordinate of the vertex is $-(-6)/(2 \cdot 1) = 6/2 = 3$, which matches the given vertex. The y-coordinate of the vertex is $f(3)$. We are given that the vertex is $(3, -5)$, so $f(3) = -5$. Substituting $x = 3$ into the function: $f(3) = (3)^2 - 6(3) + k = 9 - 18 + k = -9 + k$. Setting this equal to the given y-coordinate of the vertex: $-9 + k = -5$. Solving for k: $k = -5 + 9 = 4$. Thus, the value of the parameter k is 4. This means the function is $f(x) = x^2 - 6x + 4$. The vertex y-coordinate calculation is $f(3) = 3^2 - 6(3) + 4 = 9 - 18 + 4 = -5$. The parameter k is 4, not 1. Common algebraic traps include: 1. Errors in the vertex formula. 2. Incorrectly substituting the vertex x-coordinate back into the function. 3. Sign errors when solving for k. In this case, the correct value of k is 4.

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13 [Intermediate] • Target: Geometric series (finite)

Consider the functions $f(x) = x^2 - 6x + 5$ and $g(x) = -x + 5$. The graphs of these two functions intersect at two points. Let these points be A and B. A third function, $h(x)$, is defined as the difference between the x-coordinates of these intersection points, i.e., $h(x) = |x_A - x_B|$. What is the value of $h(x)$?

To find the intersection points of the graphs of $f(x)$ and $g(x)$, we set $f(x) = g(x)$. $x^2 - 6x + 5 = -x + 5$ Subtract $-x + 5$ from both sides: $x^2 - 6x + 5 - (-x + 5) = 0$ $x^2 - 6x + 5 + x - 5 = 0$ Combine like terms: $x^2 - 5x = 0$ Factor out x: $x(x - 5) = 0$ This gives two possible x-values for the intersection points: $x_1 = 0$ $x_2 = 5$ Let $x_A = 0$ and $x_B = 5$ (or vice versa). The problem defines $h(x)$ as the absolute difference between these x-coordinates: $h(x) = |x_A - x_B|$. $h(x) = |0 - 5| = |-5| = 5$. We are asked for the value of $h(x)$, which is 5. We need to express this value in a form that matches the choices. The choices are given as square roots. $\sqrt{16} = 4$ $\sqrt{25} = 5$ $\sqrt{36} = 6$ $\sqrt{49} = 7$ The value of $h(x)$ is 5, which corresponds to $\sqrt{25}$.

14 [Intermediate] • Target: Infinite geometric series

A bag contains 12 red marbles and 8 blue marbles. If one marble is drawn at random, what is the probability that the marble is red?

The total number of marbles in the bag is the sum of red and blue marbles: $12 + 8 = 20$ marbles. The number of favorable outcomes (drawing a red marble) is 12. The probability of an event is calculated as the ratio of the number of favorable outcomes to the total number of possible outcomes. Therefore, the probability of drawing a red marble is $P(\text{red}) = \frac{\text{Number of red marbles}}{\text{Total number of marbles}} = \frac{12}{20}$. This fraction can be simplified by dividing both the numerator and the denominator by their greatest common divisor, which is 4: $\frac{12 \div 4}{20 \div 4} = \frac{3}{5}$. Thus, the probability of drawing a red marble is $\frac{3}{5}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Advanced Arithmetic & Geometric Progressions • Cross-Lesson: 1.01 Complex Numbers

Consider a linear transformation T represented by the matrix $A = (2, 1\ 1, 3)$. Let $P = (x; y)$ be a point in the plane. The transformation maps P to P'. If the area of the region transformed by T from a unit square in the xy-plane is A, and the determinant of A is $\det(A)$, find the value of $\det(A)$ if the area of the transformed region is 5.

The area of the region transformed by a linear transformation T represented by a matrix A from a unit square is given by the absolute value of the determinant of A, i.e., Area $= |\det(A)|$. In this problem, we are given that the matrix representing the transformation is $A = (2, 1\ 1, 3)$. The determinant of A is calculated as: $\det(A) = (2 \times 3) - (1 \times 1) = 6 - 1 = 5$. We are also given that the area of the transformed region is 5. Thus, $|\det(A)| = 5$. Since we calculated $\det(A) = 5$, which is positive, the condition is satisfied. Therefore, the value of $\det(A)$ is 5.

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16 [Challenging] • Target: Summation Notation & Formulas • Cross-Lesson: 3.01 Functions & Graphs

Consider the function $f(x) = x^2 - 4x + 3$. What are the coordinates of the vertex of the parabola represented by this function?

The given function is a quadratic function in the form $f(x) = ax^2 + bx + c$, where $a = 1$, $b = -4$, and $c = 3$. The x-coordinate of the vertex of a parabola is given by the formula $x = -\frac{b}{2a}$. Substituting the values, we get $x = -\frac{-4}{2(1)} = \frac{4}{2} = 2$. To find the y-coordinate of the vertex, we substitute this x-value back into the function: $f(2) = (2)^2 - 4(2) + 3 = 4 - 8 + 3 = -1$. Therefore, the coordinates of the vertex are (2, -1).

17 [Challenging] • Target: Infinite Geometric Series & Repeating Decimals • Cross-Lesson: 4.01 Matrices & Vectors

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. Determine the y-coordinate of the vertex of the parabola.

The vertex of a parabola in the form $f(x) = ax^2 + bx + c$ occurs at $x = -\frac{b}{2a}$. In this case, $a = -1$ and $b = 4$. So, the x-coordinate of the vertex is $x = -\frac{4}{2(-1)} = -\frac{4}{-2} = 2$. To find the y-coordinate of the vertex, substitute this x-value back into the function: $f(2) = -(2)^2 + 4(2) - 3 = -4 + 8 - 3 = 1$. Therefore, the vertex is at (2, 1) and the y-coordinate is 1.

18 [Challenging] • Target: Mathematical Induction • Cross-Lesson: 5.01 Introductory Calculus

Consider the function $f(x) = x^2 - 4x + 3$. The graph of this function is a parabola. What is the y-intercept of this parabola?

The y-intercept of a function occurs where the graph crosses the y-axis, which means the x-coordinate is 0. To find the y-intercept, substitute $x = 0$ into the function $f(x) = x^2 - 4x + 3$. $f(0) = (0)^2 - 4(0) + 3$ $f(0) = 0 - 0 + 3$ $f(0) = 3$ Therefore, the y-intercept is 3. Common errors might arise from incorrectly substituting values or misinterpreting what the y-intercept represents.

19 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 6.01 Probability

The graph of a linear function $f(x)$ is shown below. What is the value of $f(0)$?

The value of $f(0)$ represents the y-intercept of the graph. On the graph, the line crosses the y-axis at the point $(0, 2)$. Therefore, $f(0) = 2$. Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

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20 [Challenging] • Target: Arithmetic series • Cross-Lesson: 1.01 Complex Numbers

Consider the matrix $A = (2, 0; 0, 3)$. When the matrix A is applied to the vector $\mathbf{v} = (x; y)$, the resulting vector is $A\mathbf{v} = (6; 9)$. What is the original vector $\mathbf{v}$?

The matrix multiplication is given by: $(2, 0; 0, 3)(x; y) = (2x; 3y)$ We are given that this result is $(6; 9)$. Therefore, we have the system of equations: $2x = 6$ $3y = 9$ Solving for x and y: $x = \frac{6}{2} = 3$ $y = \frac{9}{3} = 3$ So, the original vector $\mathbf{v}$ is $(3; 3)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Arithmetic sequences

For the arithmetic sequence with $a_1 = 0$ and common difference $d = 9$, find a_8.

a. 63 ✓ **b.** 65

c. 54 **d.** 72

$a_8 = a_1 + (8 - 1)d = 0 + 7(9) = 63$.

22 [Foundational] • Target: Arithmetic series

For the arithmetic sequence with $a_1 = -11$ and common difference $d = -5$, find a_6.

a. -36 ✓ **b.** -34

c. -41 **d.** -31

$a_6 = a_1 + (6 - 1)d = -11 + 5(-5) = -36$.

23 [Foundational] • Target: Geometric sequences

For the arithmetic sequence with $a_1 = -6$ and common difference $d = 6$, find a_7.

a. 36 **b.** 30 ✓

c. 32 **d.** 24

$a_7 = a_1 + (7 - 1)d = -6 + 6(6) = 30$.

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24 [Foundational] • Target: Geometric series (finite)

For the arithmetic sequence with $a_1 = 15$ and common difference $d = -9$, find a_7.

a. -30

b. -48

c. -37

d. -39 ✓

$a_7 = a_1 + (7-1)d = 15 + 6(-9) = -39$.

25 [Foundational] • Target: Infinite geometric series

For the arithmetic sequence with $a_1 = -6$ and common difference $d = -1$, find a_7.

a. -13

b. -11

c. -10

d. -12 ✓

$a_7 = a_1 + (7-1)d = -6 + 6(-1) = -12$.

26 [Foundational] • Target: Advanced Arithmetic & Geometric Progressions

For the arithmetic sequence with $a_1 = -3$ and common difference $d = 9$, find a_7.

a. 53

b. 42

c. 51 ✓

d. 60

$a_7 = a_1 + (7-1)d = -3 + 6(9) = 51$.

27 [Foundational] • Target: Summation Notation & Formulas

For the arithmetic sequence with $a_1 = 4$ and common difference $d = -3$, find a_{13}.

a. -30

b. -32 ✓

c. -35

d. -29

$a_{13} = a_1 + (13-1)d = 4 + 12(-3) = -32$.

28 [Foundational] • Target: Infinite Geometric Series & Repeating Decimals

Find S_6 for the arithmetic sequence with $a_1 = -13$, $d = -1$.

a. -94

b. -89

c. -97

d. -93 ✓

$a_6 = -13 + (5)(-1) = -18$. $S_6 = \frac{6(-13-18)}{2} = 3(-31) = -93$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Mathematical Induction**

Find S_4 for the arithmetic sequence with $a_1 = 2$, $d = -7$.

a. -30

b. -41

c. -38

d. -34 ✓

$a_4 = 2 + (3)(-7) = -19$. $S_4 = \frac{4(2 - 19)}{2} = 2(-17) = -34$.

30 **[Intermediate]** • **Target: Arithmetic sequences**

Find S_{10} for the arithmetic sequence with $a_1 = 3$, $d = -8$.

a. -326

b. -330 ✓

c. -338

d. -334

$a_{10} = 3 + (9)(-8) = -69$. $S_{10} = \frac{10(3 - 69)}{2} = 5(-66) = -330$.

31 **[Intermediate]** • **Target: Arithmetic series**

Find S_8 for the arithmetic sequence with $a_1 = 1$, $d = -5$.

a. -132 ✓

b. -136

c. -128

d. -137

$a_8 = 1 + (7)(-5) = -34$. $S_8 = \frac{8(1 - 34)}{2} = 4(-33) = -132$.

32 **[Intermediate]** • **Target: Geometric sequences**

Find S_{10} for the arithmetic sequence with $a_1 = 20$, $d = -7$.

a. -122

b. -115 ✓

c. -119

d. -111

$a_{10} = 20 + (9)(-7) = -43$. $S_{10} = \frac{10(20 - 43)}{2} = 5(-23) = -115$.

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33 [Intermediate] • Target: Geometric series (finite)

Find S_8 for the arithmetic sequence with $a_1 = -15$, $d = -4$.

a. -228

b. -226

c. -236

d. -232 ✓

$a_8 = -15 + (7)(-4) = -43$. $S_8 = \frac{8(-15-43)}{2} = 4(-58) = -232$.

34 [Intermediate] • Target: Infinite geometric series

Find S_{16} for the arithmetic sequence with $a_1 = -12$, $d = 3$.

a. 168 ✓

b. 164

c. 171

d. 172

$a_{16} = -12 + (15)(3) = 33$. $S_{16} = \frac{16(-12+33)}{2} = 8(21) = 168$.

35 [Intermediate] • Target: Advanced Arithmetic & Geometric Progressions

For the geometric sequence with $a_1 = -9$ and common ratio $r = -3$, find a_4.

a. 243 ✓

b. 241

c. 245

d. -243

$a_4 = a_1 r^3 = -9((-3)^3) = -9(-27) = 243$.

36 [Intermediate] • Target: Summation Notation & Formulas

For the geometric sequence with $a_1 = 9$ and common ratio $r = 3$, find a_3.

a. -81

b. 79

c. 81 ✓

d. 83

$a_3 = a_1 r^2 = 9(3^2) = 9(9) = 81$.

37 [Intermediate] • Target: Infinite Geometric Series & Repeating Decimals

Find S_4 for the geometric sequence with $a_1 = -6$, $r = 3$.

a. -240 ✓

b. -238

c. -246

d. -242

$S_4 = \frac{a_1(3^4-1)}{r-1} = \frac{-6(80)}{2} = -240$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Mathematical Induction • Cross-Lesson: 5.01 Introductory Calculus

Find S_6 for the geometric sequence with $a_1 = 8$, $r = 3$.

a. 2920

b. 2912 ✓

c. 2914

d. 2910

$S_6 = \frac{a_1(3^6 - 1)}{r - 1} = \frac{8(728)}{2} = 2912$.

39 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 6.01 Probability

Find the sum of the infinite geometric series $16 - 8 + 4 \cdots$.

a. $\frac{38}{3}$

b. $\frac{32}{3}$ ✓

c. $\frac{26}{3}$

d. $-\frac{32}{3}$

$a = 16$, $r = -\frac{1}{2}$, and $|r| < 1$, so the sum is $\frac{16}{1 - \left(-\frac{1}{2}\right)} = \frac{32}{3}$.

40 [Challenging] • Target: Arithmetic series • Cross-Lesson: 1.01 Complex Numbers

Find the sum of the infinite geometric series $144 - 24 + 4 \cdots$.

a. $\frac{878}{7}$

b. $\frac{850}{7}$

c. $\frac{864}{7}$ ✓

d. $-\frac{864}{7}$

$a = 144$, $r = -\frac{1}{6}$, and $|r| < 1$, so the sum is $\frac{144}{1 - \left(-\frac{1}{6}\right)} = \frac{864}{7}$.

41 [Challenging] • Target: Advanced Arithmetic & Geometric Progressions • Cross-Lesson: 3.01 Functions & Graphs

The sum of three consecutive terms of an arithmetic progression is 27, and their product is 504. What is the largest of the three terms?

a. 14 ✓

b. 16

c. 18

d. 21

Let terms be $a - d, a, a + d$. Sum $= 3a = 27 \Rightarrow a = 9$. Product $= (9 - d)(9)(9 + d) = 9(81 - d^2) = 504 \Rightarrow 81 - d^2 = 56 \Rightarrow d^2 = 25 \Rightarrow d = 5$. Largest term $= 9 + 5 = 14$.

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42 [Challenging] • Target: Advanced Arithmetic & Geometric Progressions • Cross-Lesson: 4.01 Matrices & Vectors

What is the sum of the series $1 + \frac{2}{3} + \frac{4}{9} + \frac{8}{27} + \ldots$?

a. 3 ✓

b. $\frac{3}{2}$

c. 2

d. $\frac{5}{2}$

$a = 1$, $r = \frac{2}{3}$. Sum $= \frac{1}{1 - 2/3} = \frac{1}{1/3} = 3$.

43 [Challenging] • Target: Summation Notation & Formulas • Cross-Lesson: 5.01 Introductory Calculus

Evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

a. 1 ✓

b. $\frac{1}{2}$

c. 2

d. Diverges

Telescoping series: $\sum(\frac{1}{n} - \frac{1}{n+1}) = 1 - 0 = 1$.

44 [Challenging] • Target: Infinite Geometric Series & Repeating Decimals • Cross-Lesson: 6.01 Probability

Express $0.\overline{36}$ as a reduced fraction.

a. $\frac{4}{11}$ ✓

b. $\frac{36}{100}$

c. $\frac{1}{3}$

d. $\frac{12}{33}$

Sum of geometric series: $\frac{36/100}{1 - 1/100} = \frac{36}{99} = \frac{4}{11}$.

45 [Challenging] • Target: Mathematical Induction • Cross-Lesson: 1.01 Complex Numbers

In proving $1 + 2 + \ldots + n = \frac{n(n+1)}{2}$, what is the inductive step?

a. **Add $(k + 1)$ to both sides of the hypothesis for $n = k$** ✓

b. Multiply both sides by 2

c. Subtract k from n

d. Differentiate with respect to n

The inductive step assumes the identity for $n = k$ and adds the next term $(k + 1)$ to prove the formula for $k + 1$.

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